

"Recent development of e-Science & Grid in Thailand"

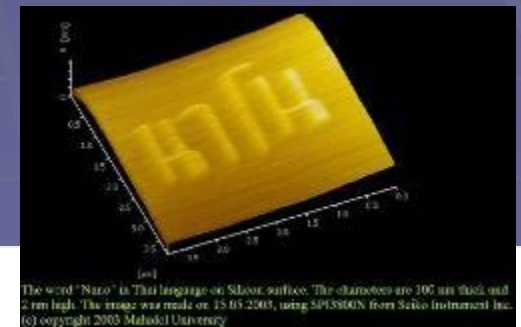
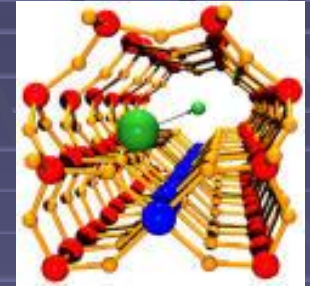
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Thailand and a New Frontier in Science and Technology

- Recently Thai Government decided to enhance Thailand competitiveness by
 - More IT investment and usage
 - Infrastructure: broadband, mobile, low-cost PC
 - E-learning, e-government
 - SIPA (Software Industry Promotion Agency)
 - Aggressive push and pursue new science
 - Excellence center in Life Science
 - Nanotechnology Center



Outline

- Definition
- Players
- Application
- Status
- Approach
- New Development

Definition

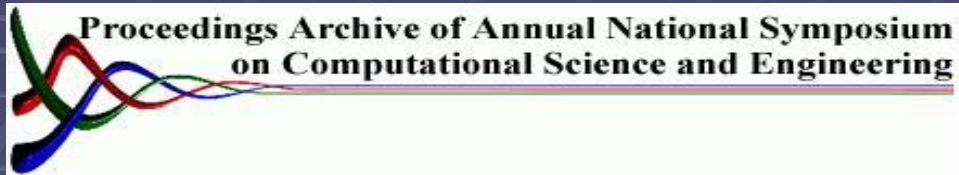
- The UK has defined e-Science as large scale sciences that will increasingly be carried out through distributed global collaborations enabled by the Internet.
- Some would say otherwise
- **“The Grid of Science”** seems to be the common adoption for Thailand

Prospect Key Players

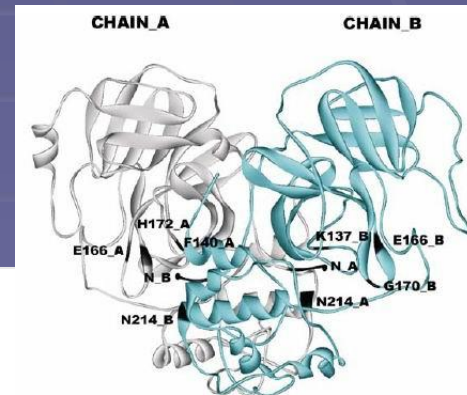
- Thaigrid → Thailand national Grid Project
- Computational Science group
- Thailand e-Science project

- A partnership project to explore grid computing technology and application in Thailand. Project started since December 2000
 - Build a grid related community for application level researchers
 - Create a grid computing infrastructure for Thai researchers
 - Stimulate the deployment of Grid Computing Technology
 - Act as a focal point for international grid collaboration
- Link: <http://www.thaigrid.net>

Community: Computational Science and Engineering Program Consortium (1995-)



- The Annual National Symposium on Computational Science and Engineering (<http://csep.hpcc.nectec.or.th/anscse/>)
 - Computational Mathematics
 - Computational Physics
 - Computational Chemistry and Biology
 - Computational Fluid Dynamics
 - Environmental Modeling etc.



Thai e-science Project (2003)

- Current members
 - Computational Chemistry Unit Cell, Department of Chemistry, Chulalongkorn University
 - Department of Computer Engineering, Chulalongkorn University
 - HPCNC, Kasetsart University
- Contact: <http://www.thai-escience.net/>
 - Dr. Prabhas Chongstitvatana (Associate Professor, Intelligent System Lab, Department of Computer Engineering, Chulalongkorn University)prabhas.c@chula.ac.th

Application Area

- Computational Fluid Dynamics
- Simulation Modeling
 - Environment
- Computational Chemistry
 - Virtual Drug Screening, Cheminformatics
- Life Science
 - Bioinformatics

Status

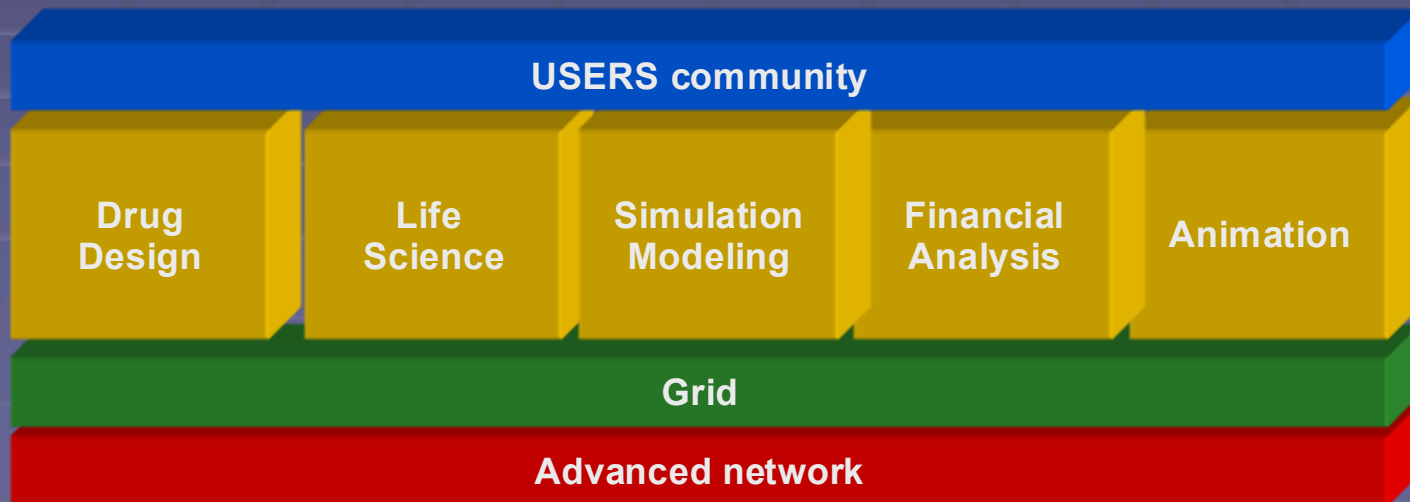
There is no “Managing e-Science” in Thailand at this point

- Missing Elements

- Need: *Appropriate* “White Paper for Thailand e-science” and *Practical* “Roadmap for Thailand e-Science” [= *strategic direction*]
- Need: Bridging Grid and Science
- Need: Coordinating function

Approach

- Build a strong scientific community (People Grid)
 - Explore innovative application including science, art, culture, entertainment
- Extend to industry by industrial partners
- Use Grid, HPC, Networking to support the mission



New Development

- TNGP can help perform coordination function to bridge "Grid" and "Science".
 - Infrastructure
 - People
 - (e-Science) Application

SEA Grid Forum

- Emerging new forum for South East Asia Collaboration
- Membership currently consists of:
 - MIMOS (Malaysia), NGO (Singapore), TNGP (Thailand)
- Details: asked Jon Lau



Thank you